

INVERTER

TEMPO DA 0 a 90%

$\alpha = \frac{S_p}{S_m}$, $\epsilon = \frac{p_m}{p_p}$, $t_r = t_f \Rightarrow p_m = p_p$, $t_p = R_{INV} C_{INV} \cdot \frac{C_L}{C_{INV}} = t_{po} \frac{C_L}{C_{INV}} \rightarrow t_{po} = R_{INV} C_{INV}$

$R_{INV} = \frac{2F(V_{DD})}{\beta_n \cdot S_{m,inv} \cdot V_{DD}}$

$C_{INV} = S_{m,inv} (1 + \alpha) C_{G1} \rightarrow C_{G1} = L_{INV}^2 C_{ox} + 2L C_{GS0}$

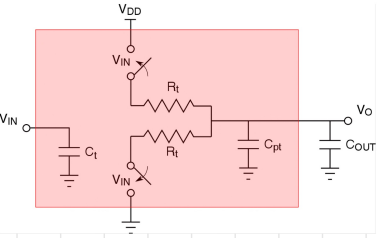
$C_{PINV} = S_{m,inv} (1 + \alpha) C_{P1} \rightarrow C_{P1} = 2L_{INV} C_{GS0} + \frac{1}{2} C_{ox} R_{DS} (L_{INV}^2 + k_{eq} L C_{G0} L_{INV})$

$F(x) = \frac{1}{1-x} \left[\frac{x}{1-x} + \frac{1}{2} \ln(1-20x) \right]$

$\frac{V_{DD}-V_T}{V_{DD}}$

dato da $t_{rx} = t_{fx}$

GATE LOGICI



$R_T = \frac{2F}{\beta_n \cdot S_{m,inv} \cdot V_{DD}}$ (con $t_{fx} = t_{rx}$)

$C_T = S_{m,gate} (1 + \alpha_{GATE}) C_{G1}$

INVERTITORI: $g_{INV} = 1$, $p_{INV} = \frac{C_{P1}}{C_{G1}}$

NAND: $g_{NAND} = \frac{m+E}{1+E}$, $p_{NAND} = m p_{INV}$, $\alpha_{NAND} = \frac{E}{m}$

NOR: $g_{NOR} = \frac{1+mE}{1+E}$, $p_{NOR} = m p_{INV}$, $\alpha_{NOR} = mE$

XOR: $g_{XOR} = m$, $p_{XOR} = 2m p_{INV}$, $\alpha_{XOR} = E$
 $g_{XOR} = m$, $p_{XOR} = 2m p_{INV}$, $\alpha_{XOR} = E$

LOGICAL EFFORT

$G = \prod_i g_i \rightarrow g = \frac{R_T C_T}{R_{INV} C_{INV}} = \frac{C_T}{C_{INV}} @ R_T = R_{INV}$

$H = \prod_i h_i = \frac{C_{OUT}}{C_{IN1}}$

$P = \sum_i p_i \rightarrow p = \frac{\sum_{i=1}^N S_i}{S_{INV}(1+E)} \cdot p_{INV}$

$B = \prod_i b_i = \prod_i \frac{C_{i+1} + C_{OFF-PATH}}{C_{i+1}}$

$F = GBH$

$\hat{f} = g_i x_i b_i = \sqrt[N]{F} \quad \forall i$

$t_{POTT} = t_{po} (N \hat{f} + P)$

1) OTTIMIZZAZIONE A NO DI STADI FISSATO

a) TRAVO $\hat{f} = \sqrt[N]{F}$

b) $\hat{f} = g_i x_i b_i = g_i b_i \cdot \frac{C_{i+1}}{C_i} \rightarrow C_i = \frac{g_i b_i C_{i+1}}{\hat{f}}$

$\Rightarrow$ si parte dall'ultimo e si arriva al 2° stadio (il 1° è fisso)

2) OTTIMIZZAZIONE A NO DI STADI VARIABILI

$t_{POTT} = t_{po} (\hat{N} \sqrt[N]{F} + \sum_{i=1}^m p_i + (\hat{N} - m) p_{INV})$

$\hat{N} = \frac{\ln F}{\ln g}$

$p \approx 0,71 p_{INV} + 2,82$ (ok per $p_{INV} < 4$)

a) $\hat{N} = \frac{\ln F}{\ln g} \rightarrow g = \hat{f} = \sqrt[N]{F} \rightarrow x_i = g_i b_i \frac{C_{i+1}}{C_i} \rightarrow C_i = \frac{g_i b_i C_{i+1}}{\hat{f}}$

(fino al 1° stadio)

b) $p \approx 0,71 p_{INV} + 2,82 \rightarrow \hat{N} = \frac{\ln F}{\ln g} \rightarrow g = \hat{f} = g_i b_i \frac{C_{i+1}}{C_i} \rightarrow C_i = \frac{g_i b_i C_{i+1}}{p}$ (fino al 2° stadio)

STADIO CON NM. DIVERSO

$x_i = g_i b_i \frac{C_{i+1}}{C_i} \rightarrow t_{us} = (\hat{N}-2) \sqrt[N]{F} + \hat{N} p + \sqrt[N]{F} + \frac{\sqrt[N]{F}}{\lambda}$

$x_{i-1} = g_i b_i \frac{C_i}{C_{i-1}}$

INTERCONNESSIONI RC

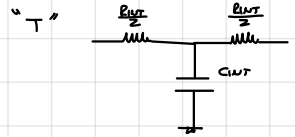
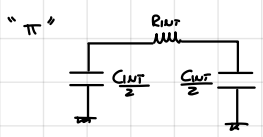
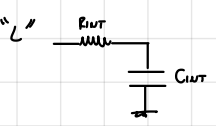
$\tau_c = \frac{\rho}{A}$, $C = C_{pp} + C_{fr} = \frac{E d_i}{t d_i} \cdot W + E d_i \cdot \frac{2\pi}{\ln \left(\frac{4 d_i + H}{H} \right)}$

$C_{fr} = \frac{E d_i 2\pi}{\cosh^{-1} \left(\frac{2 d_i + H}{H} \right)}$

T. CARATT. PAR. CONC. : $t_c = 2,3 R_{INT} C_{INT}$

T. CARATT. PAR. DISTRIBUITI : $t_d = R_{INT} C_{INT} = \frac{t_c}{2,3}$

RAPPRESENTAZIONE LINEA A PARAMETRI CONCENTRATI



$\sum_{i=1}^N R_i = R_{INT}$

$\sum_{i=1}^N C_i = C_{INT}$

ESERCIZIO

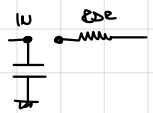
$\tau_{\lambda \rightarrow i} = \sum_{k=1}^N C_k R_{ki}$

$R_{ki} = \sum_{j=1}^N R_j \in \left\{ \begin{array}{l} \text{percorso } (\lambda \rightarrow i) \\ \text{percorso } (\lambda \rightarrow k) \end{array} \right.$

SENZA DISTRIBUIZIONI: $\tau_{0 \rightarrow u} = R_{INT} C_{INT} \cdot \frac{N(N+1)}{2 \cdot 2}$ (PER SCHEMA DA "L")

$t_{90\%} = 2,3 \tau$

DRIVER

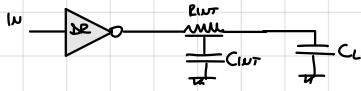


$R_{DE} = \frac{2}{\beta_n (V_{GS} - V_T)}$

$R_{DE} = \frac{R_{INV}}{2,3} = \frac{2F}{2,3 \cdot \beta_n \cdot S_{m,inv} \cdot V_{DD}}$

$R_T = R_{INV} \Rightarrow S_{m,GATE} = * \cdot S_{INV}$

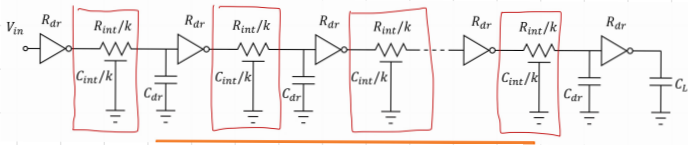
LINEA PILOTATA DA UN DRIVER



- a) $CL \neq CINT \rightarrow T_{90\%} \approx R_{INT}C_{INT} + 2,3(R_{DR}C_{INT} + R_{INT}CL + R_{DR}CL)$
- b) $CL \ll CINT \rightarrow T_{90\%} \approx R_{INT}C_{INT} + 2,3R_{DR}C_{INT}$
- c) $R_{DR} > 10 \cdot \frac{R_{INT}}{2,3} \text{ o } L_{01} < \frac{2,3R_{DR}}{10\pi} \rightarrow T_{90\%} \approx 2,3R_{DR}C_{INT}$ (LINEA CONSIDERATA COME CAPACITAN)

$\beta = \beta' \rightarrow S = \frac{W}{L} = 1$

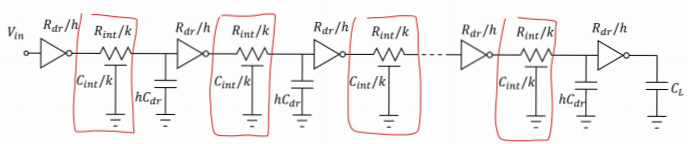
PARTIZIONAMENTO INTERCONNESSIONI



$T_{90\%} = K \left[\pi c \left(\frac{L_{01}}{K} \right)^2 + 2,3 \left(\frac{R_{DR} L_{01}}{K} + \frac{R_{INT}}{K} C_{DR} + R_{DR} C_{DR} \right) \right]$

① RIPETITORI AD ARRETA RUMINA : $K_{OTT} = \sqrt{\frac{R_{INT}C_{INT}}{2,3R_{DR}C_{DR}}}$
(se $R_{INT}C_{INT} > 2,3R_{DR}C_{INT}$)

$T_{90\%_{OTT}} = 2,3(R_{DR}C_{INT} + R_{INT}C_{DR}) + 2L_{01}\sqrt{2,3R_{DR}C_{DR}\pi c}$



$T_{90\%} = K \cdot \left[2,3 \left(\frac{R_{DR} L_{01}}{h} + \frac{R_{INT}}{h} C_{DR} + \frac{R_{DR}}{h} C_{DR} \right) + \pi c \left(\frac{L_{01}}{h} \right)^2 \right]$

② RIPETITORI A DUE OTTIPI : $h_{OTT} = \sqrt{\frac{R_{DR}C_{INT}}{R_{INT}C_{DR}}}$
(se $R_{DR}C_{INT} > R_{INT}C_{DR}$)
 K_{OTT} COME SOPRA

$T_{90\%_{DUE OTT}} \approx 7,6L_{01}\sqrt{R_{DR}C_{DR}\pi c}$

③ RIPETITORI IN CASCATA : m = numero inverteri
(utili se $2,3R_{DR}C_{INT} > R_{INT}C_{DR}$: linee considerate come capacitan CINT)

$$\begin{cases} \frac{dT_{90\%}}{dm} = 2,3R_{DR}C_{DR} - 2,3R_{DR}C_{INT} \frac{dm}{dm} \cdot q = 0 \rightarrow C_{DR} - C_{INT} \frac{dm}{dm} = 0 \\ q^{m-1} \cdot q \cdot q^m \leq q \cdot (-q^{m-1} \cdot 2,3R_{DR}C_{INT}) \\ \frac{dT_{90\%}}{dq} = 2,3(m-1)R_{DR}C_{DR} - \frac{(m-1)2,3R_{DR}C_{INT}}{q^m} = 0 \end{cases} \rightarrow q^m = \frac{C_{INT}}{C_{DR}} \rightarrow q = e$$

$T_{90\%}|_{q=e} = 2,3eR_{DR}C_{DR} \ln \left(\frac{C_{INT}}{C_{DR}} \right) + R_{INT}C_{INT}$

$T_{90\%}|_{L_{01} \text{ minima}} = 2,3R_{DR}C_{INT} + R_{INT}C_{INT}$

REDUCE IL TEMPO LEGATO AL DRIVER (SCELTA DELLA LINEA NON CORRETTA)

INTERCONNESSIONI A BASSA PERDITA

$l.c = \frac{1}{\pi \epsilon^2} = \frac{\epsilon_r \epsilon_0 \mu_0}{\pi \epsilon_0^2} \approx \frac{\epsilon_r \epsilon_0}{\pi \epsilon_0^2}$

$Z(w) = \sqrt{\frac{\pi + jwL}{g + jwC}}$

IMPERFETTA VISTA IN OGNI PUNTO DELLA LINEA LC

$z_0 = \sqrt{\frac{L}{C}} = \frac{1}{\pi \epsilon \cdot c} = \pi \epsilon \cdot l$

$t_{rl} = \frac{x}{v_z}$: temp dopo il quale trova l'ingresso nel punto x (attenuato se $x > 0$ altrimenti uguale)

LINEA PURAMENTE DISPERSIVA : $L_{01} > \frac{S Z_0}{\pi}$

LINEA A BASSA PERDITA : $L_{01} < \frac{Z_0}{2\pi}$

LINEA SENZA PERDITE : $L_{01} < \frac{2 Z_0}{\pi}$

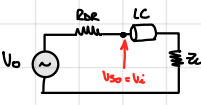
LINEA PILOTATA DA UN DRIVER

$p_L = \frac{Z_L - Z_0}{Z_L + Z_0} \rightarrow V_{TL} = V_i(1 + p_L)$
 $V_{RL} = p_L V_i$

$p_S = \frac{R_{DR} - Z_0}{R_{DR} + Z_0} \rightarrow V_{TS} = V_i(1 + p_S)$
 $V_{RS} = p_S V_i$

a) $Z_L = Z_0$ e $R_{DR} \ll Z_0 \rightarrow p_L = 0, p_S = -1, T_a = t_{rl}$

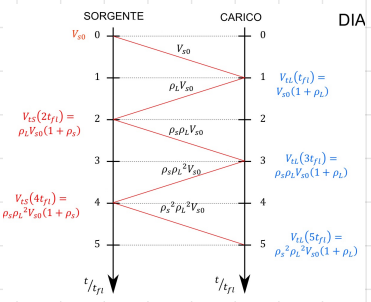
b) $R_{DR} = Z_0$ e $Z_L \gg Z_0 \rightarrow p_L = 1, p_S = 0, V_S(0) = \frac{V_0}{2} \quad T_a = t_{rl}$



$V_{S0} = V_i = \frac{Z_0}{Z_0 + R_{DR}} V_0$

temp anormale

se $t_r > 2,5 t_{rl} \Rightarrow L_{01} < \frac{t_r}{2,5 \sqrt{LC}}$ la linea può essere considerata come capacitan



$$V_S(t) = V_{S0}u(t) + \sum_{n=1}^{\infty} V_{TS}(2nt_{f1})u(t - 2nt_{f1})$$

$$= V_{S0}u(t) + \sum_{n=1}^{\infty} (1 + p_S) p_S^{n-1} \rho_L^n V_{S0}u(t - 2nt_{f1})$$

$$V_L(t) = V_{TL}(t_{f1})u(t - t_{f1}) + \sum_{n=1}^{\infty} V_{TL}(t_{f1} + 2nt_{f1})u(t - (t_{f1} + 2nt_{f1}))$$

$$= (1 + \rho_L)V_{S0}u(t - t_{f1}) + \sum_{n=1}^{\infty} (1 + \rho_L)(p_S \rho_L)^n V_{S0}u(t - (t_{f1} + 2nt_{f1}))$$

$$= (1 + \rho_L)V_{S0}u(t - t_{f1}) + (1 + \rho_L)(p_S \rho_L)V_{S0}u(t - 3t_{f1}) + (1 + \rho_L)(p_S \rho_L)^2 V_{S0}u(t - 5t_{f1}) + \dots$$